

My Sectioned Cheatsheet

1. Number Theory

1.1 Core Concepts

Def. 1 (Modular Arithmetic). Let $a, b \in \mathbb{Z}$ and $n \in \mathbb{Z}^+$. We say $a \equiv b \pmod{n}$ if $n \mid (a - b)$.

Thm. 2 (Euler’s Totient). If a and n are co-prime, then:

$$a^{\varphi(n)} \equiv 1 \pmod{n}$$

1.2 Key Proofs & Lemmas

Lem. 3 (Euclid’s Lemma). If a prime p divides ab , then $p \mid a$ or $p \mid b$.

Rem. 4. Notice the breathing room beneath the section lines now. It creates a clean visual separation without eating up too much vertical space.

Imp. 5. The section line is slightly thicker (0.8pt) than the subsection line (0.3pt).

2. Applications

2.1 Cryptography Basics

Ex. 6. Find $7^2 \pmod{5}$.
Since $7 \equiv 2 \pmod{5}$, we have:

$$7^2 \equiv 2^2 \equiv 4 \pmod{5}$$

Code Example 7

```
1 def gcd(a, b):
2     while b:
3         a, b = b, a % b
4     return a
```

Algorithm 8

```
1 int factorial(int n) {
2     if (n == 0) return 1;
3     return n * factorial(n - 1);
4 }
```

3. Miscellaneous

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Thm. 9. Notice this unlabeled block simply says "Thm. 8."

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